

Naresh Medaram

Staff Infrastructure Engineer | Distributed Systems, Cloud Networking & Reliability Engineering

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PROFESSIONAL SUMMARY

Infrastructure engineer with 11+ years setting technical direction for cloud infrastructure strategy across GCP and AWS, with deep focus on distributed systems reliability, infrastructure automation, and secure-by-default cluster networking. Built and maintained 81,000+ infrastructure-as-code files (Terraform, Ansible, Python) with zero production incidents — adopted as the organization-wide IaC standard. Architected PRISM, an 18-microservice event-driven platform reducing MTTR by 97% and MTTA by 94%. Strong systems design instinct for evaluating architectural tradeoffs across reliability, security, cost, and operational complexity — applied across cloud networking (GCP VPC, AWS multi-VPC, BGP, Cloud Armor, Cloud NAT), cluster networking (CNI/CIDR strategy, Cilium NetworkPolicy, eBPF datapath), and zero-trust security (Illumio microsegmentation, Palo Alto NGFW). Committed to Anthropic's mission: reliable, secure infrastructure is how safe AI scales.

CORE EXPERTISE

Distributed Systems & Reliability Engineering • Infrastructure as Code (Terraform, Ansible, Python) • Cluster Networking (CNI, Cilium, eBPF, NetworkPolicy, CIDR Strategy) • Cloud Networking (GCP, AWS, BGP, Cloud Armor, Cloud NAT) • Cluster Security (Zero-Trust, Least-Privilege IAM, Microsegmentation) • Systems Design Tradeoffs • Technical Direction & Cross-Team Leadership

KEY ACHIEVEMENTS

- **PRISM — 18-Microservice Distributed Incident Response Platform:** Architected end-to-end: 3-phase parallel diagnostics, automated retry-aware remediation via Event-Driven Ansible, ML-based pattern matching. MTTR: 2 hours → 72 seconds (97%). MTTA: 18 hours → 10 minutes (94%). 70% full auto-resolution. 22:1 ROI.
- **Infrastructure as Code at Scale:** 81,000+ files (Terraform 4,265 / Ansible 10,676 / Python 11,569) over 11 years, zero production incidents. Reusable Terraform modules became the organization-wide standard across 8 repositories.
- **GCP Multi-Region Architecture:** Designed and operated infrastructure across 4 global geo-locations (US, Canada, Europe, Australia) — 50+ firewalls, 100+ load balancers, multi-tenant workloads, 99.9% uptime SLA. Eliminated 15+ SPOFs with 25+ validated failover tests.
- **AWS Multi-VPC Integration:** Led integration of 40+ VPCs from a newly acquired environment within a 2-month timeline — redesigned VPC peering, Transit Gateway, Direct Connect, security groups, route tables; converted entirely to Terraform; consolidated via Alkira SD-WAN.
- **Secure-by-Default Infrastructure:** Designed and deployed Cloud Armor WAF/DDoS policies (tiered T-shirt-size policies per application) attached as backend service policies on GCP External Load Balancers; led 4-phase zero-trust macrosegmentation (500+ firewall rules, 80%+ attack surface reduction).
- **Technical Direction & Leadership:** Set infrastructure strategy and technical direction for cloud networking, IaC, and reliability engineering across US and India teams; led 125+ initiatives, drove multi-year programs to completion, mentored 40+ engineers.

TECHNICAL SKILLS

Infrastructure as Code: Terraform (modular, multi-project/region, org-wide standard), Atlantis (Terraform PR automation workflow), Ansible, Python, GitHub Actions, Jinja2, REST API automation

Distributed Systems & Orchestration: Microservices architecture, event-driven automation (Event-Driven Ansible), multi-stage workflow orchestration (GitHub Actions repository_dispatch), retry-aware remediation pipelines, systems design tradeoff evaluation (reliability vs. cost vs. operational complexity)

GCP Networking: VPC architecture (multi-project, multi-region), Cloud Interconnect (Dedicated/Partner), Cloud Router (BGP), Cloud VPN (HA VPN), Cloud Load Balancing (Internal/External/Global), Cloud NAT (port allocation tuning for high-bandwidth workloads), Cloud DNS, Cloud Armor (WAF/DDoS policies), Private Service Connect, GCVE

AWS Networking: Multi-VPC architecture, VPC peering, Transit Gateway, Direct Connect, security groups, route tables, Alkira SD-WAN

Kubernetes & Cluster Networking: CNI architecture and CIDR allocation strategy (100.64.0.0/10, per-cluster /16 blocks, dedicated pod/service/node/control-plane subnets), Cilium NetworkPolicy integration (identity-based, eBPF datapath, kube-proxy replacement), Service/Ingress design, CoreDNS with enterprise DNS, sFlow/NetFlow traffic visibility, multi-NIC awareness in hybrid cloud environments

Cluster & Host Security: Zero-trust network segmentation, Cloud Armor WAF/DDoS policies, Illumio microsegmentation (identity-based workload policy), RBAC and least-privilege IAM (GCP IAM, service account)

scoping), pod security standards and namespace isolation, Palo Alto NGFW (50+ firewalls, 4 geo-locations), IPsec/SSL VPN, DMVPN

Reliability & Observability: Grafana, Prometheus, SNMP/NetFlow/Syslog telemetry pipelines, SLO definition and alerting, production incident root-cause analysis and post-mortem-driven remediation

Routing & Switching: BGP, OSPF, EIGRP, IS-IS, BFD, route summarization & aggregation, VLAN/VTP/STP, VRRP/HSRP/GLBP

Languages & Tools: Python (11,569+ files), Terraform, Ansible, GitHub Actions, Wireshark, tcpdump

Databases: MySQL, PostgreSQL

PROFESSIONAL EXPERIENCE

Cloud Lead Network & DevOps Engineer

May 2017 – Present

Ultimate Kronos Group (UKG), Inc.

Distributed Systems & Reliability Engineering

- Architected PRISM, a Python-based, 18-microservice event-driven incident response platform — 3-phase diagnostic pipeline (parallel live health checks across 8 systems, log correlation, ML-based historical pattern matching) feeding Event-Driven Ansible for automated, retry-aware, multi-step remediation
- Result: MTTR reduced from 2 hours to 72 seconds (97%), MTTA reduced from 18 hours to 10 minutes (94%), 70% of incidents resolved with zero human intervention, 22:1 ROI on automation investment
- Eliminated 15+ single points of failure across critical load-balancing infrastructure (100% SPOF elimination) through redundant, multi-zone architecture; validated disaster-recovery readiness through 25+ documented failover tests across multiple data centers
- Led root-cause analysis for a production incident where a batch data-ingestion pipeline triggered a DNS traffic spike exhausting firewall session capacity; implemented proactive session and CPU monitoring that prevented recurrence

Infrastructure as Code — Organization-Wide Standards

- Developed and maintained 81,000+ automation files (Terraform 4,265 / Ansible 10,676 / Python 11,569) over 11 years — Python used throughout for REST API integrations, data-pipeline automation, and infrastructure tooling; reusable Terraform modules adopted as org-wide IaC standard across 8 repositories, zero production incidents
- Built subnet provisioning automation reducing turnaround from 3 days to 10 minutes (99% faster) across 320 subnets in 10 regions, with IPAM integration via Infoblox and NautoBot
- Architected multi-stage, cross-repository workflow orchestration using GitHub Actions repository_dispatch — evaluated tradeoffs between Temporal-style durable execution and event-driven Ansible remediation; selected EDA for operational simplicity and existing team expertise without sacrificing retry-aware, stateful workflow guarantees

Cloud Infrastructure & Cluster Networking

- Designed GCP infrastructure across 4 global geo-locations (US, Canada, Europe, Australia) — 50+ firewalls, 100+ load balancers, multi-tenant workloads, 99.9% uptime SLA; established Dedicated and Partner Cloud Interconnect with Cloud Router (BGP), plus Cloud VPN for redundant hybrid connectivity
- Configured Cloud NAT gateway capacity for high-bandwidth enterprise applications (Splunk, GitHub, LaunchDarkly) — minimum port allocation tuned per application based on source port consumption patterns
- Led integration of 40+ AWS VPCs from a newly acquired environment within a 2-month timeline — redesigned VPC peering, Transit Gateway, Direct Connect, security groups, and route tables; converted to Terraform; consolidated via Alkira SD-WAN
- Designed Kubernetes cluster networking architecture using the 100.64.0.0/10 address space, partitioned into per-cluster /16 blocks (supporting up to 256 clusters) with dedicated pod, service, node, and control-plane subnets — integrated CoreDNS with enterprise DNS; collaborated with platform team on Cilium CNI deployment, eBPF-based NetworkPolicy enforcement, kube-proxy replacement, and pod security namespace isolation; applied least-privilege IAM and RBAC scoping for service accounts across GKE workloads
- Architected a multi-region disaster recovery network using SD-WAN, VPC, and Private Service Connect across 4 geo-locations; managed full GCVE lifecycle (deployment 2023 through GCE migration 2026) with zero downtime

Secure-by-Default Infrastructure

- Configured Cloud Armor WAF/DDoS policies — tiered security policies (T-shirt-size configurations) attached as backend service policies on GCP External Load Balancers, tuned per application risk profile
- Led enterprise-wide zero-trust macrosegmentation across 4 phases — 500+ granular firewall rules, reducing attack surface by 80%+
- Deployed Illumio microsegmentation for identity-based, label-driven service-to-service traffic policy at the workload level — the same enforcement model as Cilium NetworkPolicy (workload identity over IP, policy-as-code), directly applicable to Kubernetes-native network security
- Managed 50+ Palo Alto firewalls across 4 global geo-locations with proactive Grafana/Prometheus monitoring — pre-emptively prevented CPU exhaustion and session-limit incidents through telemetry-driven alerting

Observability & Incident Response

- Built network observability and telemetry pipelines — SNMP, NetFlow, and Syslog feeding Grafana and Prometheus dashboards — defined SLOs for network availability with proactive alerting before threshold breaches
- Performed production troubleshooting using packet-level analysis (Wireshark, tcpdump) across multi-vendor and containerized (GKE) environments; post-mortem-driven systemic remediation standard across the team
- Engineered two internal Python platforms from design to production: a RAG-based self-service chatbot (60% reduction in documentation search time) and a natural-language network query API exposing 13 firewall-intelligence tools at sub-100ms latency — both adopted by 40+ engineers, eliminating tribal knowledge bottlenecks

Cross-Team Leadership & Engineering Growth

- Led 125+ technical initiatives across US and India engineering teams, consistently driving multi-year programs to completion
- Mentored 40+ engineers through self-service platforms, documentation, and technical coaching; drove GCP resource labeling initiative establishing cost-governance standards adopted by the FinOps team

Cloud Network Engineer

July 2015 – May 2017

Comtek Global, Inc.

- Designed and deployed multi-vendor network infrastructure (Cisco, Juniper, F5, Palo Alto) — routers, switches, firewalls, load balancers across enterprise and WAN environments
- Configured BGP, OSPF routing and established site-to-site VPN tunnels; deployed F5 LTM/GTM with custom iRules and load-balancing algorithms
- Managed SSL certificate lifecycle for customer-facing applications (DigiCert, GoDaddy, Comodo)

Graduate/Teaching Assistant

August 2014 – May 2015

University of Houston - Clear Lake, TX

- Supported 45 graduate students in network engineering coursework and program accreditation
- Managed Cisco Prime, Infoblox (DNS/DHCP/IPAM), and Cisco ASA firewalls; created MOPs for OS upgrades and Layer 2/3 protocol configuration

Network Engineer / Network Administrator

May 2010 – November 2013

Devblaze Pvt. Ltd., Levonsys Pvt. Ltd., Global Research Solutions

- Built foundational multi-vendor routing, switching, and firewall expertise (Cisco Nexus/ASR/Catalyst, Juniper SRX) — OSPF, EIGRP, BGP, VLAN/STP, Layer 2 security, vPC, VDC, FCoE, F5 LTM/GTM
- Led Cisco-to-Juniper migration: configuration conversion, SRX firewall cluster creation, redundancy testing, MOP execution

PROFESSIONAL DEVELOPMENT & INDUSTRY ENGAGEMENT

- **Google Cloud Next 2025 — Global AI & Cloud Conference (2025):** Engaged with Google Cloud engineering leaders on generative AI infrastructure, multi-cloud architecture, and enterprise AI adoption patterns
- **ONUG AI Summit, New York, NY (2024):** Participated in Open Networking User Group summit focused on AI-driven network operations and open networking standards for enterprise IT
- **FLNUG AI Summit, Tampa, FL (March 2025):** Invited by Network to Code to serve as a panelist on applied AI for network engineering, automation, and operations
- **Juniper Networks Industry Event, Miami, FL (2025):** Engaged with Juniper Networks roadmap sessions on AI-driven networking, automation, and enterprise security

EDUCATION

Master of Science, Computer Engineering

University of Houston-Clear Lake, Houston, Texas | 2015 | GPA 3.7/4.0

Bachelor of Engineering, Electronics and Communication Engineering

DRK Institute of Science and Technology, Andhra Pradesh, India | 2013 | GPA 81.83%

CERTIFICATIONS

- GCP Professional Cloud Network Engineer
- Terraform Associate (HashiCorp)
- Palo Alto Certified Network Security Engineer (PCNSE)
- Cisco Certified Network Professional (CCNP)
- Juniper Networks: JNCIP-ENT, JNCIS-ENT, JNCIA
- F5 Certified Administrator, BIG-IP | Red Hat Certified Specialist in Ansible Automation (EX407) | OpenStack Certified Administrator (COA) | GCP Cloud Digital Leader, GCP Gen AI Leader, Claude Code Architect